Q11. Comparative Analysis of Zero-Shot, One-Shot, and Few-Shot Prompting

Aim: To compare zero-shot, one-shot, and few-shot prompting techniques based on response quality.

Algorithm:

1. Define a task to explain the benefits of regular exercise.
2. Create zero-shot, one-shot, and few-shot prompts.
3. Prepare sample responses for each technique.
4. Evaluate the responses based on relevance, accuracy, completeness, and clarity.
5. Calculate the average score and identify the best technique.

Program

```
# Sample responses for comparison
responses = {
    "Zero-Shot": (
        "Exercise improves health, helps control weight, "
        "and reduces stress."
    ),
    "One-Shot": (
        "Regular exercise strengthens muscles, supports heart health, "
        "and improves mood."
    ),
    "Few-Shot": (
        "1. Physical health: Strengthens muscles and the heart.\n"
        "2. Weight management: Burns calories and supports a healthy weight.\n"
        "3. Mental health: Reduces stress and improves mood."
    )
}

# Illustrative scores: Relevance, Accuracy, Completeness, Clarity
scores = {
    "Zero-Shot": [4, 5, 4, 4],
    "One-Shot": [4, 5, 4, 4],
    "Few-Shot": [5, 5, 5, 5]
}

print("Task:", task)

for method in responses:
    average = sum(scores[method]) / len(scores[method])
    print("\nTechnique:", method)
    print("Response:", responses[method])
    print("Average Score:", round(average, 2), "/ 5")

best = max(scores, key=lambda x: sum(scores[x]) / len(scores[x]))
print("\nBest Technique:", best)
```

Output:

```
Task: Explain three benefits of regular exercise.

Technique: Zero-Shot
Response: Exercise improves health, helps control weight, and reduces stress.
Average Score: 4.25 / 5

Technique: One-Shot
Response: Regular exercise strengthens muscles, supports heart health, and improves mood.
Average Score: 4.25 / 5

Technique: Few-Shot
Response: 1. Physical health: Strengthens muscles and the heart.
2. Weight management: Burns calories and supports a healthy weight.
3. Mental health: Reduces stress and improves mood.
Average Score: 5.0 / 5

Best Technique: Few-Shot
```

Result: Few-shot prompting achieved the highest illustrative score because examples help guide the response's structure and level of detail.